COLUMBIA UNIVERSITY
IN THE CITY OF NEW YORK

STAT 5703

Statistical Inference and Modeling

Dobrin Marchev

Hypothesis Tests
Part II

From Last Time

If the sample is normally distributed, then:

$$\boxed{\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}}$$

Corollary: Recall from STAT 5701 that $E(\chi^2_\nu) = \nu$

$$\Rightarrow E\left[\frac{(n-1)S^2}{\sigma^2}\right] = E(\chi^2_{n-1}) = n-1$$

$$\Leftrightarrow \frac{(n-1)E(S^2)}{\sigma^2} = n-1$$

$$\Leftrightarrow E(S^2) = \sigma^2$$

That is, this shows S^2 is an unbiased estimator for σ^2. However, this proof assumes normal sample.

Example 1

Let $X_1, X_2, \ldots, X_n$ be a random sample from the population random variable $X \sim \mathrm{N}(\mu, \sigma)$

Test: H_0: $\sigma \leq 5$ vs. H_a: $\sigma > 5$

Using the rejection region

$$R = \left\{ \sum_{i=1}^{n} (X_i - \bar{X})^2 > c \right\}$$

With $n = 45$, $s = 6.42$ and $\alpha = 0.05$

a) Find c;

b) State the conclusion.

Example 1 Solution

We have that:

$$\frac{(n-1)s^2}{\sigma^2} \sim X_{n-1}^2, \text{where } s^2 = \sum \frac{(x_i - \bar{x})^2}{n-1}$$

$$\Rightarrow \pi(\sigma) = P(R|\sigma) = \mathrm{P}(\textstyle\sum_{i=1}^{n}(X_i - \bar{X})^2 > c|\sigma)$$

$$= P\left(\frac{(n-1)S^2}{\sigma^2} > \frac{c}{\sigma^2}\right) = P\left(X_{n-1}^2 > \frac{c}{\sigma^2}\right)$$

$$= 1 - F_{\chi_{n-1}^2}\left(\frac{c}{\sigma^2}\right)$$

where $F_{\chi_{n-1}^2}$ is the cdf of the χ_{n-1}^2 distribution.

Note that $\pi'(\sigma) = -f_{\chi_{n-1}^2}\left(\frac{c}{\sigma^2}\right)(-2c)\sigma^{-3} = \frac{2c}{\sigma} f_{\chi_{n-1}^2}\left(\frac{c}{\sigma^2}\right) > 0$

Therefore, $\pi(\sigma)$ is an increasing function.

Example 1 Solution

$$\Rightarrow \max_{H_0} \pi(\sigma) = \max_{\sigma \le 5} \pi(\sigma) = \pi(5)$$

$$\Rightarrow \alpha = \pi(5) \Leftrightarrow 0.05 = 1 - F_{\chi^2_{44}}\left(\frac{c}{25}\right)$$

$$\Leftrightarrow F_{\chi^2_{44}}\left(\frac{c}{25}\right) = 0.95$$

$$\Leftrightarrow \frac{c}{25} = 95^{th} \text{ quatile of } \chi^2_{44} = 60.481$$

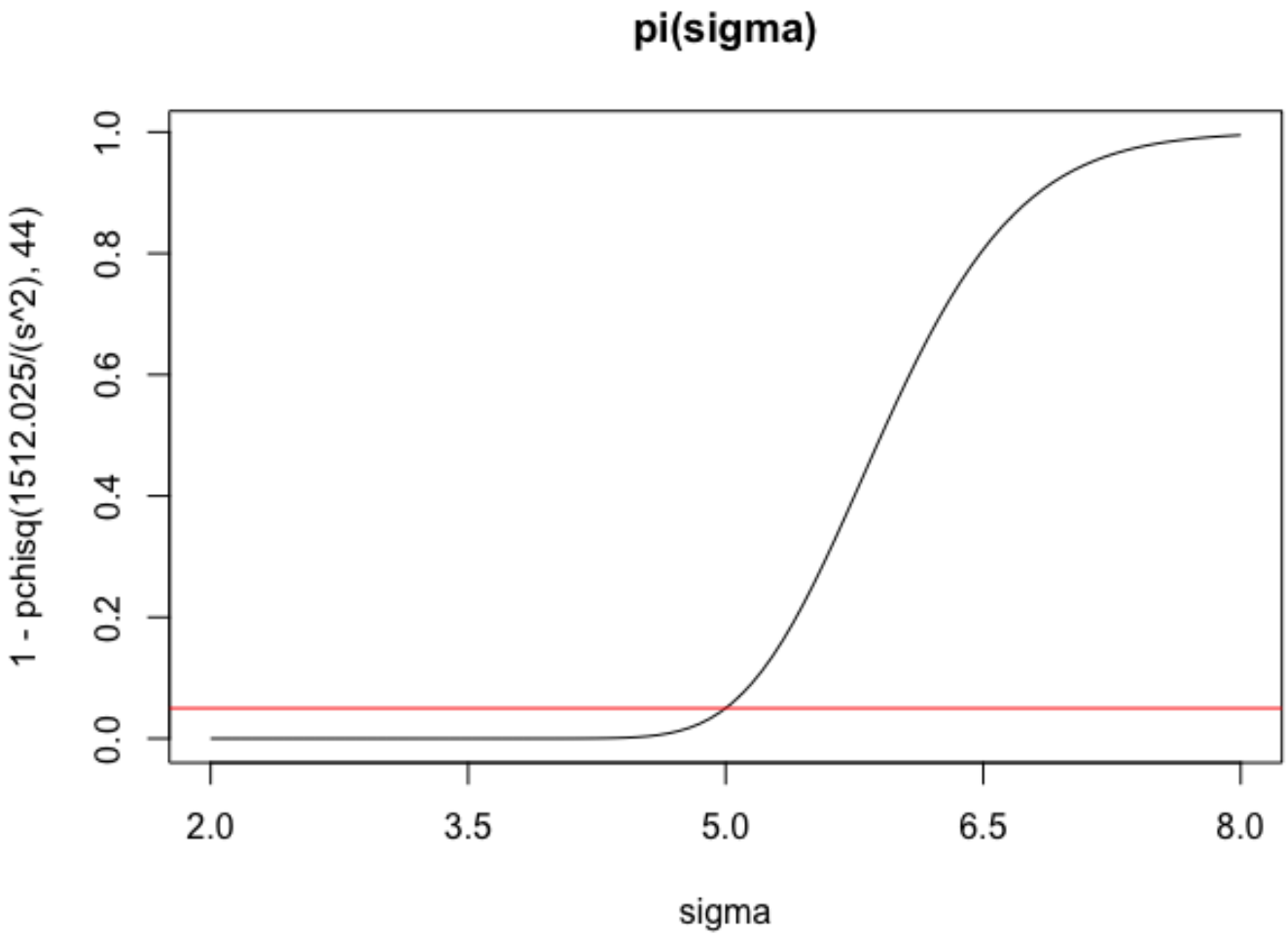
(from table)

$$\Rightarrow c = 25(60.481) = \boxed{1512.025}$$

b) $\sum (x_i - \bar{x})^2 = (n-1)s^2 = 44(6.42)^2 = 1813.52$

Since $1813.52 > 1512.025$ we reject H_0

Example 1: Graph of $\pi(\sigma)$



Recall: Small sample *t*-test for μ when σ is unknown

Assumption: $X_1, X_2, \ldots, X_n \sim \mathrm{N}(\mu, \sigma)$

Hypotheses:

	Left-tail	Two-tail	Right-tail
H_0	$\mu \geq \mu_0$	$\mu = \mu_0$	$\mu \leq \mu_0$
H_a	$\mu < \mu_0$	$\mu \neq \mu_0$	$\mu > \mu_0$
RR	$T < -t_\alpha$	$\lvert T \rvert > t_{\alpha/2}$	$T > t_\alpha$

Test Statistic:

$$T = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

Question: Why is $T \sim t_{n\text{-}1}$?

t-test statistic

We have: $Z = \frac{\bar{X}-\mu_0}{\frac{\sigma}{\sqrt{n}}} \sim N(0,1)$ & $U = \frac{(n-1)s^2}{\sigma^2} \sim \chi^2_{n-1}$

Also, $\bar{X}$ and s^2 are independent (by Basu's Theorem).

Then, by the definition of *t*-distribution: $\frac{Z}{\sqrt{\frac{U}{n-1}}} \sim t_{n-1}$

But

$$\frac{Z}{\sqrt{\frac{U}{n-1}}} = \frac{\frac{\frac{\bar{X}-\mu_0}{\sigma}}{\sqrt{n}}}{\sqrt{\frac{\frac{(n-1)s^2}{\sigma^2}}{n-1}}} = \frac{\sqrt{n}\frac{\bar{X}-\mu_0}{\sigma}}{\sqrt{\frac{(n-1)s^2}{(n-1)\sigma^2}}} = \frac{\bar{X}-\mu_0}{\frac{s}{\sqrt{n}}} = T$$

Wald Test

Suppose we want to test a general hypothesis:

$$H_0: \theta = \theta_0 \text{ vs. } H_a: \theta \neq \theta_0$$

Let $\hat{\theta}_{\text{MLE}}$ be the MLE of θ, and $\widehat{SE}(\hat{\theta})$ be the *estimated* SE of $\hat{\theta}$. Assume the regularity conditions hold so MLE is asymptotically normally distributed.

Then the test statistic is

$$W = \frac{\hat{\theta} - \theta_0}{\widehat{SE}(\hat{\theta})} \xrightarrow{d} N(0,1)$$

Note 1: Sometimes the test statistic is defined as W^2 and then the reference distribution is χ^2_1

Note 2: If $SE(\hat{\theta})$ can be evaluated when $\theta = \theta_0$, then it is better to use $\text{SE}(\theta_0)$, instead of $\widehat{SE}(\hat{\theta})$.

Example 2

Let

$$X \sim \text{Binom}(n, p)$$

Test:

$$\text{H}_0\text{: } p = p_0 \text{ vs. } \text{H}_a\text{: } p \neq p_0$$

Parameter: θ is p

MLE : $\hat{\theta}_{MLE}$ is $\frac{X}{n}$ = "sample proportion" = $\hat{p}$

SE : $\sqrt{Var(\hat{\theta})} = \sqrt{Var\left(\frac{X}{n}\right)} = \sqrt{\frac{Var(X)}{n^2}}$

$$= \sqrt{\frac{np(1-p)}{n^2}} = \frac{\sqrt{p(1-p)}}{\sqrt{n}}$$

Example 2: Solution

$$SE(\hat{p}) = \sqrt{\frac{p(1-p)}{n}} \Rightarrow \widehat{SE}(\hat{p}) = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

$$\Rightarrow W = \frac{\hat{\theta} - \theta_0}{\widehat{SE}} = \boxed{\frac{\hat{p} - p_0}{\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}} \approx N(0,1)}$$

Wald z-test for population proportion

Z-test for Population Proportion

Let

$$X \sim \text{Binom}(n, p)$$

Hypotheses:

	Left-tail	Two-tail	Right-tail
H_0	$p \geq p_0$	$p = p_0$	$p \leq p_0$
H_a	$p < p_0$	$p \neq p_0$	$p > p_0$
RR	$Z < -z_\alpha$	$\lvert Z \rvert > z_{\alpha/2}$	$Z > z_\alpha$

But

$$Z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1 - p_0)}{n}}}$$

Note the switch to p_0 in the denominator!

Example 3:

Test:

$$H_0: p \leq 0.3 \text{ vs. } H_a: p > 0.3$$

a) Given: $n = 4115$, $x = 1276$, $\alpha = 0.1$, find the test statistic and state the conclusion.

b) Find the power of the test when $p = 0.4$

Example 3 (solution):

$$\hat{p} = \frac{1276}{4115} = 0.31$$

$$\Rightarrow Z = \frac{\hat{p} - p_o}{\sqrt{\frac{p_0(1-p_0)}{n}}} = \frac{0.31 - 0.3}{\sqrt{\frac{0.3(0.7)}{4115}}} = 1.4$$

Significance level $\alpha = 0.1 \Rightarrow z_\alpha = 1.28$

Conclusion: Since 1.4 > 1.28, we reject H_0

Therefore, we conclude $p > 0.3$

Example 3 (continued): Find the power of the test when $p = 0.4$

$$\pi(p) = P(R \mid p)$$

When $p = 0.4 \Rightarrow E(\hat{p}) = 0.4$, $Var(\hat{p}) = \frac{0.4(0.6)}{4115}$

$$\Rightarrow \pi(0.4) = P(Z > z_\alpha \mid p = 0.4)$$

$$= P\left(\frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}} > z_\alpha \mid p = 0.4\right)$$

$$= P\left(\hat{p} > p_0 + z_\alpha \sqrt{\frac{p_0(1-p_0)}{n}} \mid p = 0.4\right)$$

Example 3 (cont.):

$$\pi(0.4) = P\left(\hat{p} > p_0 + z_\alpha \sqrt{\frac{p_0(1-p_0)}{n}} \,\middle|\, p = 0.4\right)$$

$$= P\left(\frac{\hat{p} - 0.4}{\sqrt{\frac{0.4(0.6)}{4115}}} > \frac{p_0 + z_\alpha \sqrt{\frac{p_0(1-p_0)}{n}} - 0.4}{\sqrt{\frac{0.4(0.6)}{4115}}}\right)$$

$$= P\left(Z > \frac{0.3 + 1.28\sqrt{\frac{0.3(0.7)}{4115}} - 0.4}{\sqrt{\frac{0.4(0.6)}{4115}}}\right)$$

$$= P(Z > -11.9) \approx 1$$

Aside:

Power of the z-test for H_a: $p > p_0$

Let $p_1 > p_0$

Then $$\pi(p_1) = 1 - \Phi\left[\frac{p_0 - p_1 + z_\alpha\sqrt{\frac{p_0(1-p_0)}{n}}}{\sqrt{\frac{p_1(1-p_1)}{n}}}\right]$$

cdf of $N(0,1)$

Exercise 1

Suppose we have two independent samples as follows:

$$X \sim \text{Binom}(n_1, p_1) \ \& \ Y \sim \text{Binom}(n_2, p_2)$$

Test:

$$\text{H}_0\text{: } p_1 = p_2 \text{ vs. } \text{H}_a\text{: } p_1 \neq p_2$$

Hint: The hypotheses are equivalent to

$$\text{H}_0\text{: } p_1 - p_2 = 0 \text{ vs. } \text{H}_a\text{: } p_1 - p_2 \neq 0$$

P-value

Definition: The *p*-value of a hypothesis test is the probability of obtaining a value of the test statistic which is *more extreme* than the one observed in the current sample, assuming H_0 is true.

More extreme means more contradictory to H_0

Let X be the test statistic (as r.v./formula) and x be its observed value (when calculated from the sample). Then:

- If right-tail H_a: $\theta > \theta_0$, then *p*-value = $P(X > x \mid H_0)$
- If left-tail H_a: $\theta < \theta_0$, then *p*-value = $P(X < x \mid H_0)$
- If two-tail H_a: $\theta \neq \theta_0$, then *p*-value = $P(|X| > |x| \mid H_0)$

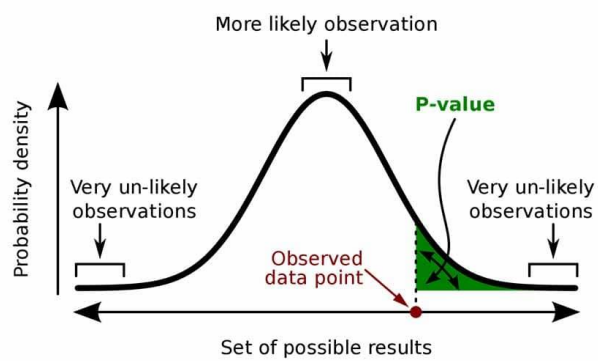
Example 3 (cont.): Find the p-value

$H_a: p > 0.3 \rightarrow$ right-tail

$\Rightarrow$ p-value $= P(Z > 1.4 \mid H_0)$

$= 1 - 0.9192 = \boxed{0.0808}$

Conclusion:
Since p-value = 0.0808 < 0.1 = α,
we reject H_0



A **p-value** (shaded green area) is the probability of an observed (or more extreme) result assuming that the null hypothesis is true.

How do we use the *p*–value?

Fundamental Rule of Statistic:

Reject H_0 if p-value $< \alpha$

That's why the p-value is aka the lowest α level at which H_0 can be rejected.

Example 5: Exercise

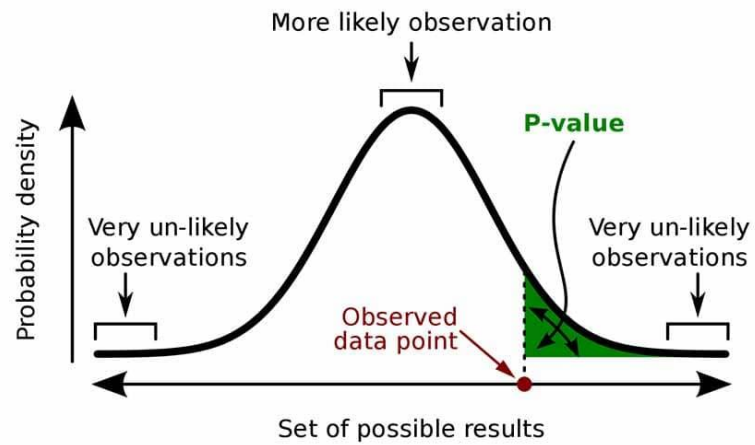
Test the hypotheses:

H_0: $\mu \leq 2$ vs. H_a: $\mu > 2$

Given

$$\bar{X} = 2.06, s = 0.141, n = 51, \alpha = 0.05$$

Find the p-value



A **p-value** (shaded green area) is the probability of an observed (or more extreme) result assuming that the null hypothesis is true.

Example 6

H_0: $\mu \geq 15$ versus H_a: $\mu < 15$

with a sample of 10 observations.

Test statistic was $t = -2.77$

Find the p-value with R.

Solution:

```
pt(-2.77, 9)
[1] 0.01087675
```

Example 7:

A movie theater owner likes to keep a very close eye on how much popcorn goes into each bag sold, so he knows that the amount is normally distributed and the average bag has 8 cups of popcorn and that this varies a little bit, about half a cup. That is, the population mean should be $\mu = 8.00$ and the known population standard deviation is $\sigma = 0.50$. The owner wants to make sure that the newest employee is filling bags correctly, so over the course of a week he randomly assesses 25 bags filled by the employee to test for a difference ($n = 25$). He doesn't want bags overfilled or under filled, so he looks for differences in both directions.

Example 7:

Hypotheses:

$$H_0: \mu = 8 \; vs. \; H_a: \mu \neq 8$$

Data:

7.6, 7.4, 7.9, 7.4, 7.7, 8.2, 8.2, 8.1, 7.8, 7.8, 8.3, 7.7, 7.9, 7.1, 7.6, 7.8, 7.8, 8.8, 6.8, 7.8, 8.4, 6.7, 6.3, 7.2, 7.3

$$\bar{x} = 7.664, n = 25$$

Test statistic:

$$Z_{test} = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{7.664 - 8}{0.5/\sqrt{25}} = -3.36$$

P-value:

$$P(|Z| > |-3.36|) = 2P(Z < -3.36)$$
$$= 2(0.0004) = 0.0008$$

Example 7:

What if we drop the info σ = 0.50?

Then this becomes a t-test.

Data:

$$\bar{x} = 7.664, s = 0.56, n = 25$$

Test statistic:

$$T_{test} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} = \frac{7.664 - 8}{0.56/\sqrt{25}} = -3$$

P-value:

$$P(|T_{24}| > |-3|) = 2P(T_{24} < -3)$$

Use software: `2*pt(-3, 24) = 0.006205737`

Example 7:

```
import numpy as np
from scipy.stats import ttest_1samp

data = np.array([7.6, 7.4, 7.9, 7.4, 7.7, 8.2, 8.2, 8.1, 7.8,
 7.8, 8.3, 7.7, 7.9, 7.1, 7.6, 7.8, 7.8, 8.8, 6.8, 7.8, 8.4,
 6.7, 6.3, 7.2, 7.3
])
population_mean = 8

t_stat, p_value = ttest_1samp(data, population_mean)

# Output results
print(f"t-statistic: {t_stat}")
print(f"P-value: {p_value}")

statistic: -2.9938011536722677
P-value: 0.006297782945388144 5
```

Hypothesis Tests
Neyman-Pearson Theorem

The theory works only for *simple* hypotheses.

Definition:

A simple hypothesis is of the form H: $\theta = \theta_0$

Neyman – Pearson Theorem

Theorem (Neyman – Pearson): For testing

$$H_0: \theta = \theta_0 \text{ vs. } H_a: \theta = \theta_a$$

Let $R = \left\{(x_1, \ldots, x_n): \frac{L(\theta_a)}{L(\theta_0)} \geq k\right\}$, where

$L(\theta) = f(x_1, \ldots, x_n; \theta)$ is the likelihood function and k is such that $P(R \mid H_0) = \alpha^*$. Then any other test with level $\alpha \leq \alpha^*$ will be such that $\beta^* \leq \beta$. In other words, the rejection region R corresponds to the *most powerful* test.

Example 8

Let

$$X_1, \ldots, X_n \sim \text{Bernoulli}(p)$$

a) Find the RR of the most powerful test for testing

$$H_0: p = ½ \text{ vs. } H_a: p = ¼$$

b) If $n = 8$, find c such that α^* is at most 0.05

Recall: When $X \sim \text{Bernoulli}(p)$, then

$$p(x) = p^x(1-p)^{1-x}$$

Example 8 (Answers)

Let

$$X_1, \ldots, X_n \sim \text{Bernoulli}(p)$$

a) Find the RR of the most powerful test for testing

$$H_0: p = ½ \text{ vs. } H_a: p = ¼$$

$$R = \left\{(x_1, \ldots, x_n): \sum_{i=1}^{n} X_i \leq c\right\}$$

b) If $n = 8$, find c such that α^* is at most 0.05

Solution: $c = 1$, since

```
pbinom(0:8,8,0.5)
[1] 0.00390625 0.03515625 0.14453125 0.36328125 0.63671875
[6] 0.85546875 0.96484375 0.99609375 1.00000000
```

Example 9

Consider iid sample

$$X_1, \ldots, X_n \sim \text{Poisson}(\lambda)$$

a) Find the RR of the most powerful test for testing

$$H_0: \lambda = 1 \text{ vs. } H_a: \lambda = 2$$

b) If $n = 5$, $c = 10$, find α^*

Example 9 (Solution)

Let

$$X_1, \ldots, X_n \sim \text{Poisson}(\lambda)$$

Find the RR of the most powerful test for testing

$$H_0: \lambda = 1 \text{ vs. } H_a: \lambda = 2$$

The pmf of Poisson is $f(x;\lambda) = \frac{\lambda^x e^{-\lambda}}{x!}$

$$\Rightarrow L(\lambda) = \prod_{i=1}^{n} \left(\frac{\lambda^{x_i}}{x_i!} \boldsymbol{e}^{-\lambda} \right) = \frac{\lambda^{\sum x_i} \mathbf{e}^{-n\lambda}}{\prod_{i=1}^{n} x_i!}$$

Example 9: Solution

$$\Rightarrow L(\lambda_0) = L(1) = \frac{1^{\sum x_i}\mathbf{e}^{-n(1)}}{\prod_{i=1}^{n} x_i!} = \frac{\mathbf{e}^{-n}}{\prod_{i=1}^{n} x_i!}$$

$$L(\lambda_a) = L(2) = \frac{2^{\sum x_i}\mathbf{e}^{-n(2)}}{\prod_{i=1}^{n} x_i!}$$

$$\Rightarrow \frac{L(\lambda_a)}{L(\lambda_0)} = \frac{\dfrac{2^{\sum x_i}\mathbf{e}^{-n(2)}}{\prod_{i=1}^{n} x_i!}}{\dfrac{\mathbf{e}^{-n}}{\prod_{i=1}^{n} x_i!}} = \boldsymbol{e}^{-n}2^{\sum x_i}$$

$$\Rightarrow \frac{L(\lambda_a)}{L(\lambda_0)} \geq k \Leftrightarrow 2^{\Sigma x_i} \geq ke^n \Leftrightarrow \sum_{i=1}^{n} X_i \geq \log_2 ke^n = c$$

Example 9 Solution

$$\Rightarrow \quad RR: \left\{ x_1, \dots, x_n : \boxed{\sum_{i=1}^{n} x_i \geq c} \right\}$$

---

b) $n = 5$, $c = 10$, $\alpha^* = ?$

$$\alpha^* = P(RR \mid H_0) = P\left(\underbrace{\sum_{i=1}^{5} x_i}_{\sim Pois(5)} \geq 10 \;\middle|\; x_i \sim Pois(1) \right)$$

$$= \underbrace{1 - \text{ppois}(9, 5)}_{\text{in } R} = 0.032 = \alpha^*$$

$$1 - P(Y \leq 9) = P(Y \geq 10)$$

Example 10 (Exercise)

Let

$$X_1, \ldots, X_n \sim \text{Exp}(\lambda)$$

Find the RR of the most powerful test for testing

$$H_0: \lambda = 1 \text{ vs. } H_a: \lambda = 2$$

Example 11

Let

$$X_1, \ldots, X_n \sim \mathrm{N}(\mu, 1)$$

Find the RR of the most powerful test for testing

$$\mathrm{H}_0: \mu = \mu_0 \text{ vs. } \mathrm{H}_a: \mu = \mu_1$$

Solution:

$$f(x;\mu) = \frac{1}{\sqrt{2\pi}}\mathbf{e}^{-\frac{(x-\mu)^2}{2}} = (2\pi)^{-\frac{1}{2}}\mathbf{e}^{-\frac{1}{2}(x-\mu)^2}$$

$$\Rightarrow L(\mu) = \prod_{i=1}^{n}[(2\pi)^{-\frac{1}{2}}\mathbf{e}^{-\frac{1}{2}(x_i-\mu)^2}]$$

$$= (2\pi)^{-\frac{n}{2}}\mathbf{e}^{-\frac{1}{2}\sum_{i=1}^{n}(x_i-\mu)^2}$$

Example 11 (solution):

$$L(\mu) = (2\pi)^{-\frac{n}{2}}\mathbf{e}^{-\frac{1}{2}\sum_{i=1}^{n}(x_i-\mu)^2}$$

$$\Rightarrow \frac{L(\mu_1)}{L(\mu_0)} = \frac{(2\pi)^{-\frac{n}{2}}\mathbf{e}^{-\frac{1}{2}\sum_{i=1}^{n}(x_i-\mu_1)^2}}{(2\pi)^{-\frac{n}{2}}\mathbf{e}^{-\frac{1}{2}\sum_{i=1}^{n}(x_i-\mu_0)^2}}$$

$$= \mathbf{e}^{-\frac{1}{2}\left[\sum_{i=1}^{n}(x_i-\mu_1)^2-\sum_{i=1}^{n}(x_i-\mu_0)^2\right]}$$

$$= \mathbf{e}^{-\frac{1}{2}\left[\sum_{i=1}^{n}\left(x_i-2\mu_1 x_i+\mu_1^2\right)-\sum_{i=1}^{n}\left(x_i-2\mu_0 x_i+\mu_0^2\right)\right]}$$

$$= \mathbf{e}^{-\frac{1}{2}\left(-2\mu_1\sum_{i=1}^{n}x_i+n\mu_1^2+2\mu_0\sum_{i=1}^{n}x_i+n\mu_0^2\right)}$$

Example 11 (continued)

$$= e^{-\frac{1}{2}\left[2(\mu_0-\mu_1)\sum x_i + n\mu_1^2 - n\mu_0^2\right]} \geq k$$

$$\Longleftrightarrow \quad -\frac{1}{2}\left[2(\mu_0-\mu_1)\sum x_i + n\mu_1^2 - n\mu_0^2\right] \geq k_1$$

$$(\mu_1-\mu_0)\sum x_i - \frac{1}{2}(n\mu_1^2 - n\mu_0^2) \geq k_1$$

$$\Longleftrightarrow \quad \frac{(\mu_1-\mu_0)\sum x_i}{\mu_1-\mu_0} \geq \frac{k_2}{\mu_1-\mu_0}$$

when $\mu_1 > \mu_0$

Example 11 (Solution):

Assume, w.o.l.g. that $\mu_1 > \mu_0$. Then the rejection region is

$$\boldsymbol{RR}: \frac{\sum_{i=1}^{n} X_i}{n} \geq \frac{c}{n} \Leftrightarrow \bar{X} \geq c_1$$

Now, assume $\alpha = 0.05, n = 10$. Let's find c_1.
Under the null hypothesis,

$$\bar{X} \sim N\left(\mu_0, \frac{\sigma}{n}\right) = N\left(\mu_0, \frac{1}{n}\right)$$

We must have that $P(RR|H_0) = \alpha$

$$\Leftrightarrow P(\bar{X} \geq c_1 | \mu = \mu_0) = \mathrm{P}\left(Z \geq \frac{c_1 - \mu_0}{\frac{1}{\sqrt{n}}}\right) = 0.05$$

$$\Leftrightarrow \frac{c_1 - \mu_0}{\frac{1}{\sqrt{n}}} = 1.645 \Leftrightarrow c_1 = \mu_0 + 1.645 \times \frac{1}{\sqrt{n}}$$

Example 12

Let

$$X_1, \ldots, X_n \sim \mathrm{N}(0, \sigma)$$

Find the RR of the most powerful test for testing

$$\mathrm{H}_0: \sigma^2 = \sigma^2{}_0 \text{ vs. } \mathrm{H}_a: \sigma^2 = \sigma^2{}_a$$

where $\sigma^2{}_0 < \sigma^2{}_a$

$$f(x;\sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{x^2}{2\sigma^2}} = (2\pi)^{-\frac{1}{2}} (\sigma^2)^{-\frac{1}{2}} e^{-\frac{1}{2}\frac{x^2}{\sigma^2}}$$

Example 12 (Solution):

$$L(\sigma^2) = \prod_{i=1}^{n} [(2\pi)^{-½}(\sigma^2)^{-½} e^{-\frac{x_i^2}{2\sigma^2}}$$

$$= (2\pi)^{-\frac{n}{2}}(\sigma^2)^{-\frac{n}{2}} e^{-\frac{1}{2\sigma^2}\sum_{i=1}^{n} x_i^2}$$

$$\Rightarrow \frac{L(\sigma_a^2)}{L(\sigma_0^2)} = \frac{(2\pi)^{-\frac{n}{2}}(\sigma_a^2)^{-\frac{n}{2}} e^{-\frac{1}{2\sigma_a^2}\sum_{i=1}^{n} x_i^2}}{(2\pi)^{-\frac{n}{2}}(\sigma_0^2)^{-\frac{n}{2}} e^{-\frac{1}{2\sigma_0^2}\sum_{i=1}^{n} x_i^2}}$$

$$= (\sigma_0^2)^{\frac{n}{2}}(\sigma_a^2)^{-\frac{n}{2}} e^{\frac{1}{2\sigma_0^2}\sum_{i=1}^{n} x_i^2 - \frac{1}{2\sigma_a^2}\sum_{i=1}^{n} x_i^2}$$

Finish as an exercise

Exercise: Example 14

Let $X_1, \ldots, X_n$ be a random sample from a population with pdf

$$f(x) = \theta x^{\theta-1}, \qquad 0 < x < 1$$

a) Find the RR of the most powerful test for testing

$$H_0: \theta = 3 \text{ vs. } H_a: \theta = 2$$

b) Find the critical value c if $n = 1$, $\alpha = 0.05$